
Name:

Entry No.:

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1. [2 marks] Suppose that ϕ and ψ are formulas such that $\phi \models \psi$. Show that if ϕ and ψ have no variable in common then either ϕ is unsatisfiable or ψ is valid.

Ans: We will show that if ψ is not valid, then ϕ is not satisfiable. On the contrary, suppose that ψ is not valid but ϕ is satisfiable. This means we have an assignment A_1 that satisfies $\neg\psi$, and another assignment A_2 that satisfies ϕ . Since A_1 and A_2 do not have any variables in common, we can combine them into a single assignment A . For any variable v , A copies the assignment that v has in A_1 or A_2 (this is possible without a conflict because we know that v appears in only one of A_1, A_2).

Now the assignment A makes ψ false (because A and A_1 are the same as far as the variables in ψ are concerned), and it makes ϕ true (once again, because A and A_2 are the same as far as the variables in ϕ are concerned). This contradicts that $\phi \models \psi$, which essentially says that any assignment that makes ϕ true must make ψ true.

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2. [2 marks] Give a natural deduction proof of the law of excluded middle using basic proof rules.

Ans:

1	$\neg(\phi \vee \neg\phi)$	assumption
2	ϕ	assumption
3	$\phi \vee \neg\phi$	$\vee i_1$ 2
4	$\perp$	$\neg e$ 3, 1
5	$\neg\phi$	$\neg i$ 2–4
6	$\phi \vee \neg\phi$	$\vee i_2$ 5
7	$\perp$	$\neg e$ 6, 1
8	$\neg\neg(\phi \vee \neg\phi)$	$\neg i$ 1–7
9	$\phi \vee \neg\phi$	$\neg\neg e$ 8

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3. [1 marks] Is it true that we can transform any (natural deduction) proof of $\phi_1, \phi_2, \dots, \phi_n \vdash \psi$ into a proof of the theorem $\vdash \phi_1 \rightarrow (\phi_2 \rightarrow (\phi_3 \rightarrow (\dots \rightarrow (\phi_n \rightarrow \psi) \dots)))$? If not, explain why. If yes, sketch the transformation.

Ans: The given statement is true. The proof of $\phi_1, \phi_2, \dots, \phi_n \vdash \psi$ can be transformed in the following steps: i) change the *premises* into *assumptions*, and open n boxes (as you do in case of assumptions), and ii) now close each box one by one, using the rule $\rightarrow_i$, starting from ϕ_n (to get $\phi_n \rightarrow \psi$), all the way to ϕ_1 (to get $(\phi_2 \rightarrow (\phi_3 \rightarrow (\dots \rightarrow (\phi_n \rightarrow \psi) \dots)))$).

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4. [5 marks] Recall that if we have clauses $(p \vee q)$ and $(r \vee \neg q)$, then we can apply resolution to get the clause $(p \vee r)$. Does this mean that the formula $(p \vee q) \wedge (r \vee \neg q)$ is equivalent to $(p \vee r)$? In particular:
- (a) [2 marks] Is $(p \vee q) \wedge (r \vee \neg q) \vdash (p \vee r)$ valid? If so, give a natural deduction proof for this. If not, justify your answer using a suitable valuation.

Ans: This sequent is valid. Here is a natural deduction proof.

1. $(p \vee q) \wedge (r \vee \neg q)$	<i>premise</i>
2. $p \vee q$	$\wedge_{e1}$ 1
3. p	<i>assume</i>
4. $p \vee r$	$\vee_{i1}$ 3
5. q	<i>assume</i>
6. $(r \vee \neg q)$	$\wedge_{e2}$ 1
7. r	<i>assume</i>
8. $p \vee r$	$\vee_{i2}$ 7
9. $\neg q$	<i>assume</i>
10. $\perp$	$\neg_e$ 5, 9
11. $p \vee r$	$\perp_e$ 10
12. $p \vee r$	$\vee_e$ 6, 7 – 8, 9 – 11
13. $p \vee r$	$\vee_e$ 2, 3 – 4, 5 – 12

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- (b) **[2 marks]** Is $(p \vee r) \vdash (p \vee q) \wedge (r \vee \neg q)$ valid? If so, give a natural deduction proof for this. If not, justify your answer using a suitable valuation.

Ans: No, this sequent is not valid. The valuation $\langle p \mapsto T, q \mapsto T, r \mapsto F \rangle$ makes $(p \vee r)$ true, but falsifies $(p \vee q) \wedge (r \vee \neg q)$.

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- (c) [1 marks] Two formulas are said to be *equisatisfiable* if the first formula is satisfiable whenever the second is and vice versa; in other words, either both formulas are satisfiable or neither is. The truth values of two equisatisfiable formulas may nevertheless disagree for a particular assignment of variables.

Are the two formulas $((p \vee q) \wedge (r \vee \neg q))$, and $(p \vee r)$ equisatisfiable? Why or why not?

Ans: Yes, the two formulas are equisatisfiable.

One direction is trivial. We have proved above that $(p \vee q) \wedge (r \vee \neg q) \vdash (p \vee r)$ valid. Therefore, $(p \vee q) \wedge (r \vee \neg q) \models (p \vee r)$. Clearly, any valuation that makes $(p \vee q) \wedge (r \vee \neg q)$ true, also makes $(p \vee r)$ true.

For the other direction, assume $(p \vee r)$ is satisfiable. Then there must be a satisfying valuation v . If v makes both p and r true, then $(p \vee q) \wedge (r \vee \neg q)$ is satisfiable. If not, it must make exactly one of them true. If it makes p true, we set $v' = \{v; q \mapsto \text{false}\}$, and if it makes r true then we set $v' = \{v; q \mapsto \text{true}\}$. Clearly, v' is a satisfying assignment of $(p \vee q) \wedge (r \vee \neg q)$.

(It also suffices to simply show that both the formulas are satisfiable.)

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